

ECON 21030: Honors Econometrics

Regression In Practice

Spring Quarter 2024

Regression in the sample

In our last lecture, we covered population linear regression. This entire discussion defined coefficients from these linear models in terms of characteristics of the joint distribution of random variables. Now we're going to discuss how we can try to learn about population regression coefficients in sample.

In particular, let's assume we're interested in population regression:

$$Y = X'\beta + U$$

where Y is an outcome and X is a k -dimensional vector of regressors. You can assume that one component of X (typically the first one) is equal to one, so that the model includes a constant, although this is not important. We know that $\beta = E[XX']^{-1}E[XY]$. But this is an estimand—a feature of distributions that exists (for all practical purposes in theory only). How can we estimate it?

With data, of course. Assume we have access to an iid sample (Y_i, X_i) of size N , where $X_i = (X_{i1}, \dots, X_{ik})'$. The population linear regression tells us that $Y_i = X_i'\beta + U_i$, where u_i is the residual for observation i . The trouble of course is that we don't know β , so we'll need an estimator.

The most popular estimator of β is the **ordinary least squares** (OLS) estimator, which we will denote $\hat{\beta}$. This estimator is constructed by trying to do what the β does in the population—minimize residual sum of squares—in the sample:

$$\hat{\beta} = \min_{\beta} \frac{1}{N} \sum_{i=1}^N (Y_i - X_i'\beta)^2$$

if you attempt to solve this problem, you will find that $\hat{\beta}$ is defined by:

$$0 = \sum_{i=1}^N 2X_i(Y_i - X_i'\beta)$$

which implies a unique values for $\hat{\beta}$ given by

$$\hat{\beta} = \left(\frac{1}{N} \sum_i X_i X_i' \right)^{-1} \frac{1}{N} \sum_i X_i Y_i$$

whenever $\left(\frac{1}{N} \sum_i^N X_i X_i'\right)$ can be inverted.

Notice how close this is to the population formula for β . $\left(\frac{1}{N} \sum_i^N X_i X_i'\right)$ looks like a good estimator for $E[XX']$. And $\frac{1}{N} \sum_i^N X_i Y_i$ looks like a good estimator for $E[XY]$. This is a general idea in econometrics known as the [plug-in principle](#). Sometimes estimators of composite estimands can be constructed using the [sample analogues](#) of their constituent parts.

Notice that we could have also derived $\hat{\beta}$ using the sample analogue of the other population restrictions that define β . For example, requiring that the sample mean of $X_i \hat{U}_i = X_i(Y_i - X_i' \hat{\beta})$ is zero implies that:

$$\sum_{i=1}^N X_i(Y_i - X_i' \hat{\beta}) = 0$$

when X_i includes a constant this constraint is naturally going to produce the same estimator of β (if not, we also need to impose that the sample mean of $\hat{U}_i$ is zero). Estimators constructed to satisfy a set of sample moment conditions are known as [generalized method of moments](#) estimators, a very powerful tool you will encounter in future econometric work.

Dealing with all of this summation notation can be annoying. It's very common to stack the sample of outcomes into a n -by-1 vector of outcomes $\mathbf{Y}$, stack the regressors into n -by- k matrix of covariates $\mathbf{X}$, sometimes called a [design matrix](#) because it might be directly specified by the researcher, and stack the residuals into a n -by-1 vector $\mathbf{U}$. Notice that this means each *row* i of $\mathbf{X}$ is a the transposition of the random vector of covariates for observation i , where as normally we think of X_i as a column vector. Written this way, we can define write the regression equation in vector form as:

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{U}$$

And we can define the OLS estimator $\hat{\beta}$ as:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

Be sure you understand the difference in how we write projection, where we have $X'\beta$, and this sample version, where we use $\mathbf{X}\beta$. You should also convince yourself, as an [exercise](#) that $(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$ is algebraically equivalent to the version written with summation nation.

Why is $\hat{\beta}$ a good estimator of β ? First, note that by the weak law of large numbers $\frac{1}{N} \mathbf{X}'\mathbf{X} \xrightarrow{p} E[XX']$ whenever it applies element-wise to this matrix, which requires $Var(X_k X_j) < \infty$ for all k, j . Likewise $\frac{1}{N} \mathbf{X}'\mathbf{Y} \xrightarrow{p} E[XY]$ whenever $Var(X_k Y) < \infty$ for all k . Continuous mapping tells us that $\left(\frac{1}{N} \mathbf{X}'\mathbf{X}\right)^{-1} \xrightarrow{p} E[XX']^{-1}$ whenever the inverse exists at $E[XX']$. Finally, Slutsky tells us that $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y} \xrightarrow{p} \beta$. So we know that as we collect more data, $\hat{\beta}$ converges in probability to β . That is good!

When is $\hat{\beta}$ also unbiased for β ? Consider that:

$$\begin{aligned} E[\hat{\beta}] &= E[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}] \\ &= E[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'(\mathbf{X}\beta + \mathbf{U})] \\ &= \beta + E[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{U}] \end{aligned}$$

β is defined such that $E[XU] = 0$. Unfortunately, this is not enough to guarantee that $\hat{\beta}$ is unbiased. In the bivariate regression case, for example, any bias in OLS is given by the $E[\widehat{Cov}(X, U)/\widehat{Var}(X)]$, where the wide hat notation indicates sample covariances/variances. Although $E[\widehat{Cov}(X, U)] = 0$, the expectation of a ratio of random variables is not in general equal to the ratio of their expectations, and in this case correlation between $\widehat{Cov}(X, U)$ and $\widehat{Var}(X)$ can lead to some bias. Your second problem set is going to walk you through an example of this issue.

For OLS to be unbiased, therefore, we need a stronger condition. One sufficient condition is that mean U is independent of X , which implies that $E[U|X] = 0$. Simple iterated expectations then shows that:

$$E[\hat{\beta}] = \beta + E[E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U}|\mathbf{X}]] = \beta$$

When $E[U|X] = 0$, we know that $E[Y|X] = X'\beta$, which implies the CEF is linear. OLS is therefore unbiased when the CEF is linear. In practice, however, we tend not too worry too much about the bias of OLS for the population projection coefficient even when the CEF is potentially nonlinear because we still know it is a) consistent, and b) OLS is semi-parametrically efficient, which means, loosely, that it is not possible to construct a better estimator without further assumptions. Note also that the fact that because $E[U|X] = 0$ holds for any $\mathbf{X}$, this means that OLS is not only unbiased unconditionally but also holding fixed a set of regressors $\mathbf{X}$.

Regression mechanics

The sample analogue motivation for $\hat{\beta}$ implies that some characteristics of β that hold in the *population* will hold for $\hat{\beta}$ in the *sample*. As we discussed above, for example, $\hat{\beta}$ is the coefficient vector that makes $\sum_{i=1}^N X_i \hat{U}_i = 0$. This holds *exactly* in the sample of N observations we have, not in expectation. Since X_i includes a constant, it is also true that $\sum_{i=1}^N \hat{U}_i = 0$ *exactly*.

You can see this directly using the definition of $\hat{\beta}$, since:

$$\begin{aligned}\mathbf{X}'\hat{\mathbf{U}} &= \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\beta}) \\ &= \mathbf{X}'(\mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}) \\ &= \mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{Y} = 0\end{aligned}$$

As a result, the correlation between X and $\hat{U}$ will be exactly zero in sample, as well the correlation between linear functions of X such as $X'\hat{\beta}$ and $\hat{U}$. In this sense OLS *overfits* the data, since we know that for the true population regression coefficient β it is not true that in any given sample $\mathbf{X}'\mathbf{U} = 0$ (this only holds in expectation). Nevertheless, in its efforts to minimize the *sample* sum of squares, OLS always guarantees that $\mathbf{X}'\hat{\mathbf{U}} = 0$ exactly.

This implies that in-sample fits are going to exceed the fit of the population regression. In particular, the sample R^2 given by:

$$R^2(\hat{\beta}) = 1 - \frac{\hat{\mathbf{U}}'\hat{\mathbf{U}}}{(\mathbf{Y} - \bar{Y})(\mathbf{Y} - \bar{Y})} = 1 - \frac{\widehat{Var}(\hat{U})}{\widehat{Var}(Y)}$$

is generally going to *overestimate* the population R^2 because we know that $Var(\hat{U})$ is generally smaller than $Var(U)$. This is true even when the CEF is linear, so that OLS is an unbiased estimator of both the population regression coefficient and the CEF. In particular, adding an additional regressor to a regression can *never* decrease the sample R^2 , even if this regressor's population coefficient is exactly zero. For this reason many statistical packages report an [adjusted \$R^2\$](#) that accounts for the number of regressors in the model in an effort to mitigate some of this overfitting problem.

The R^2 described above is known as the [centered \$R^2\$](#) or the [coefficient of determination](#). If the regressors included only a constant, then $Var(\hat{U})$ would simply be the sample mean of $(Y_i - \bar{Y})^2$, which is an estimate of the variance of Y . Hence the R^2 would be zero. One can also compute the [uncentered \$R^2\$](#) , which is simply one minus the ratio of residual to explained sum of squares: $\hat{Y}'\hat{Y}/Y'Y = 1 - \hat{U}'\hat{U}/Y'Y$. Sometimes the latter will make more sense as a measure of fit in models where a constant is not appropriate.

Studying the finite sample properties of OLS is somewhat easier with some notation for the matrices above. We call $P_X = X(X'X)^{-1}X'$ the [projection matrix](#) for X because it *projects* whatever it is applied to onto the column space of X . Thus $P_X Y = \hat{Y}$, for example, the sample fits. The [annihilator matrix](#) or [residual-maker](#) is related and defined as $M_X = I - P_X$, where I is the n -by- n identity matrix. Thus $M_X Y$ gives the OLS residuals.

These matrices have a handful of useful properties. First, you can verify that they are *square* and *symmetric*. They are also [idempotent](#), which means that $AA = A$. For example:

$$P_X P_X = X(X'X)^{-1}X'X(X'X)^{-1}X' = P_X$$

The two matrices are also [orthogonal](#), which means that $P_X M_X = 0$, and you can verify as an [exercise](#). These two properties are intuitive from the geometric perspective: once we've projected a vector onto the column space of X , projecting again should not affect the fits. Likewise, M_X projects vectors into the space orthogonal to X , so projecting once is the same as projecting twice. Naturally the projection of a vector that lives in the space orthogonal to X back onto X is going to give us the null space, so $P_X M_X = 0$.

We can use the properties of these matrices to prove that [Frisch-Waugh-Lovell](#) also holds exactly in sample. Consider a partition of the regressors into $[X \ W]$ and the sample regression:

$$Y = X\hat{\beta} + W\hat{\gamma} + \hat{U}$$

Calling this the “sample regression” makes it explicit that $X'\hat{U}$ and $W'\hat{U}$ both equal zero exactly in sample. If we were to residualize both sides with respect to W , we would have:

$$\begin{aligned} M_W Y &= M_W X\hat{\beta} + M_W W\hat{\gamma} + M_W \hat{U} \\ \tilde{Y} &= \tilde{X}\hat{\beta} + \hat{U} \end{aligned}$$

where $\tilde{Z} = M_W Z$. Why? Note that:

$$\begin{aligned} M_W W &= (I - P_W) W = W - W(W'W)^{-1}W'W = 0 \\ M_W \hat{U} &= (I - P_W) \hat{U} = \hat{U} - W(W'W)^{-1}W'\hat{U} = \hat{U} \end{aligned}$$

Notice that $\tilde{\mathbf{X}}'\hat{\mathbf{U}} = \mathbf{X}'\mathbf{M}_{\mathbf{W}}'\hat{\mathbf{U}} = \mathbf{X}'\hat{\mathbf{U}} = 0$. So $\hat{\beta}$ is also the least squares solution to the regression of $\tilde{\mathbf{Y}}$ onto $\tilde{\mathbf{X}}$. You actually already knew this, since $\hat{\mathbf{U}}$ is in-sample orthogonal to $\mathbf{X}$ and $\mathbf{W}$, and $\tilde{\mathbf{X}}$ is a linear combination of $\mathbf{X}$ and $\mathbf{W}$.

One can also show that it is unnecessary to residualize $\mathbf{Y}$ by considering:

$$\begin{aligned}\mathbf{Y} &= \tilde{\mathbf{X}}\hat{\beta} + \hat{\mathbf{V}} \\ \hat{\mathbf{V}} &= \hat{\mathbf{U}} + \mathbf{Y} - \tilde{\mathbf{Y}}\end{aligned}$$

Here we know that:

$$\begin{aligned}\tilde{\mathbf{X}}'(\hat{\mathbf{U}} + \mathbf{Y} - \tilde{\mathbf{Y}}) &= \tilde{\mathbf{X}}'(\mathbf{Y} - \tilde{\mathbf{Y}}) \\ &= \mathbf{X}'\mathbf{M}_{\mathbf{W}}(\mathbf{Y} - \mathbf{M}_{\mathbf{W}}\mathbf{Y}) \\ &= \mathbf{X}'\mathbf{M}_{\mathbf{W}}(\mathbf{I} - \mathbf{P}_{\mathbf{W}})\mathbf{Y} = 0\end{aligned}$$

Thus $\hat{\beta}$ is also the solution to the regression of $\mathbf{Y}$ onto $\tilde{\mathbf{X}}$. Intuitively, leaving the component of $\mathbf{Y}$ that is a linear combination of $\mathbf{W}$ in $\mathbf{Y}$ on the left-hand side is harmless because $\tilde{\mathbf{X}}$ is already orthogonal to $\mathbf{W}$. Whether or not you residualize $\mathbf{Y}$, however, is important for standard errors, as we will discuss later.

Regression inference under normality

Now that we know how to estimate β and understand the properties of $\hat{\beta}$, let's turn to inference. How can we quantify our uncertainty about β ? How can we test whether β is consistent with certain hypotheses such as $\beta = 0$?

As in our review of stats, the first approach we used was to make enough assumptions that we know the finite-sample distribution of our estimator. We could, for example, invoke normality. In fact, the linear regression model was traditionally studied under the assumption that Y is normally distributed given X . One can show that if $Y|X \sim N(X'\beta, \sigma^2)$ OLS is the [maximum likelihood estimator](#) (MLE) of β .¹ MLEs are another class of estimators that we won't touch on too deeply here but you will encounter frequently in future econometric coursework. Though modern methods don't lean too heavily on the normal motivation for OLS, there are enough hangovers in popular statistical packages and testing methods that its worth reviewing this model in detail.

Notice that when $Y|X \sim N(X'\beta, \sigma^2)$ we can write $Y = X'\beta + e$, where $e|X \sim N(0, \sigma^2)$. The normal model thus implies a linear conditional expectation function and a [homoskedastic error](#), which means that the residual variance is independent of X . The residual has both a conditional and marginal distribution that is governed by one parameter (σ). Since observations are iid, realizations of e are mutually independent. This implies the random vector $\mathbf{e}$ has conditional distribution $\mathbf{e}|X \sim N(\mathbf{0}, \sigma^2\mathbf{I}_N)$, where $\mathbf{I}_N$ is the N -by- N identity matrix.

In the normal model, the sampling distribution of the OLS estimator is thus given by $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}$, which is a linear combination of normal random variables, making $\hat{\beta}$ itself normal

¹Recall that if (Y, X) are jointly normal, then the conditional distribution of Y given X is normal with a mean that is linear in X . That is a slightly stronger assumption than what is made in what I am calling the normal regression model, since X can be non-normal (i.e., discrete).

as well. Conditional on $\mathbf{X}$, its variance is given by:

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)' | \mathbf{X}] &= E[(\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{e}\mathbf{e}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} | \mathbf{X}] \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

By conditioning on $\mathbf{X}$ here, you might think I am throwing out some uncertainty if the design matrix is itself random (you can easily imagine cases where it is not because it is specified by the researcher / statistician). If it is random, then the law of total variance tells us that $\text{Var}(\hat{\beta})$ is $E[\text{Var}(\hat{\beta} | \mathbf{X})] + \text{Var}(E[\hat{\beta} | \mathbf{X}])$. But we know that under the assumption of the normal regression model, $E[\hat{\beta} | \mathbf{X}] = \beta$, so the second part is zero, while the first part is just $\sigma^2 E[(\mathbf{X}'\mathbf{X})^{-1}]$. The unconditional variance of $\hat{\beta}$ is thus just the average of the conditional variance.

Putting things together, we have that $\hat{\beta} - \beta | \mathbf{X} \sim N(0, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$. This is a description of the sampling variation of the entire *vector* $\hat{\beta}$. Any individual element k is marginally normally distributed with variance given by the k th diagonal of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

Inference and hypothesis testing are now straightforward. Following the steps developed in our previous lectures, for example, one can test hypotheses such as $\beta_k = 0$ and construct confidence intervals for each element. The standard way to do the former is through the use of a **z-score**. For the null that $\beta_k = b$, the z-score is given by:

$$z_k = \frac{\hat{\beta}_k - b}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}_{kk}}}$$

where the kk subscript indicates we are taking the element on the k th diagonal. The z-score is standard normally distributed under the null, making it straightforward to construct an α -level test or confidence intervals with $1 - \alpha$ coverage. You may be concerned by the fact that we are still conditioning on $\mathbf{X}$. If the testing procedure has size α conditional on $\mathbf{X}$, however, it must the same size unconditionally as well. I leave the simple proof of this claim as an **exercise**.

You may also be concerned about the fact the z-score depends on the potentially unknown σ^2 . In settings like this, parameters such as σ^2 are often referred to as **nuisance parameters**—a quantity that is not intrinsically interesting, but must be known to estimate or conduct inference on the parameters we do care about, such as β . In this case, it is natural to replace σ^2 with its sample estimator: $\hat{\sigma}^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(N - K)$, where $\hat{\mathbf{e}} = \mathbf{Y} - \mathbf{X}\hat{\beta}$ and K is the **degrees of freedom**, typically the number of regressors in $\mathbf{X}$. This adjustment makes $\hat{\sigma}^2$ unbiased because:

$$\begin{aligned} E[\hat{\mathbf{e}}'\hat{\mathbf{e}} | \mathbf{X}] &= E[\mathbf{e}'\mathbf{M}_\mathbf{X}\mathbf{e} | \mathbf{X}] \\ &= \sigma^2 \text{trace}(\mathbf{M}_\mathbf{X}) = \sigma^2(N - K) \end{aligned}$$

where the first line follows from the fact that $\mathbf{M}_\mathbf{X}\mathbf{e} = \mathbf{M}_\mathbf{X}(\mathbf{Y} - \mathbf{X}\beta) = \hat{\mathbf{e}}$ and the fact that $\mathbf{M}_\mathbf{X}$ is idempotent and symmetric. The last line follows because for a random vector X with mean μ and covariance matrix Σ , $E[X'AX] = \mu'A\mu + \text{trace}(A\Sigma)$, and:

$$\begin{aligned} \text{trace}(\mathbf{M}_\mathbf{X}) &= \text{trace}(\mathbf{I}_N) - \text{trace}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\ &= N - \text{trace}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) = N - K \end{aligned}$$

where the last line follows because for conformable matrices AB , $\text{trace}(AB) = \text{trace}(BA)$. You have in fact already seen a version of this where $K = 1$ in the unbiased estimator of a sample variance (i.e., a regression where $\mathbf{X}$ only includes the constant term).

Plugging in $\hat{\sigma}^2$ (and continuing to condition on $\mathbf{X}$), we have:

$$t_k = \frac{\hat{\beta}_k - b}{\sqrt{\hat{\sigma}^2(\mathbf{X}'\mathbf{X})_{kk}^{-1}}}$$

This object has a t -distribution with $N - K$ degrees of freedom, since:

$$t_k = \frac{\hat{\beta}_k - b}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})_{kk}^{-1}}} \sqrt{\frac{\sigma^2}{\hat{\sigma}^2}} = \frac{z_k}{\sqrt{\frac{\hat{\mathbf{e}}'\hat{\mathbf{e}}/(N-K)}{\sigma^2}}}$$

The numerator is standard normal, while the denominator is independent $\chi^2(N - k)/(N - K)$.² As discussed in our lectures on probability, this random variable has a t -distribution. Even when we have to estimate σ^2 , therefore, we still know the sampling distribution of $\hat{\beta}$ and can conduct hypothesis tests using **t-stats**, construct p -values using the CDF of the t -distribution, and invert hypothesis tests to construct confidence intervals. This test statistic does not depend on any nuisance parameters. In fact it is **pivotal**, meaning that its distribution does not depend on any unknowns. Notice also that as N becomes very large, the t distribution converges to a standard normal. This reflects the fact that $\hat{\sigma}^2 \xrightarrow{P} \sigma^2$, so with enough data the t stat is going to start looking a lot like the z -score.³ When n is smaller, however, t -based inference is going to be more conservative (i.e., less likely to reject the null) because the t distribution has fatter tails and thus larger critical values.

You can also see that the t -stat is directly connected to the **standard error** for $\hat{\beta}_{kk}$, which is simply the k th diagonal of the *estimator* of the variance of $\hat{\beta}$, i.e. $\widehat{Var}(\hat{\beta}|\mathbf{X}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$. The t -stat for the null that $\beta_k = 0$ is just the ratio of $\hat{\beta}_k$ to the square root of its standard error. This is the t -stat that will be reported in most statistical packages' implementation of OLS by default. A $1 - \alpha$ confidence interval will be symmetric in this case around $\hat{\beta}_k$ and given by:

$$[\hat{\beta}_k - SE(\hat{\beta}_k)t_{\alpha/2}(N - K), \hat{\beta}_k + SE(\hat{\beta}_k)t_{\alpha/2}(N - K)]$$

where $t_{\alpha/2}(N - K)$ are critical values and $SE(\hat{\beta}_k)$ is the standard error for $\hat{\beta}_k$. It is very common to report $SE(\hat{\beta}_k)$ along with $\hat{\beta}$ so people can construct these types of confidence intervals in their heads. Many people rely on a rule of thumb that 5% critical value for a t stat is about 2, so 95% confidence intervals are just $\hat{\beta}_k \pm 2 \cdot SE(\hat{\beta}_k)$.

We may also be interested certain **linear hypothesis** that involve combinations of elements of β . We may wish to know, for example, whether the coefficients on any two regressors are the same or whether a set of coefficients are jointly equal to zero. We can write these hypotheses as:

$$H_0 : R\beta = r$$

where R is an m -by- k matrix and r is an m by one vector. The row dimension of R , m , is the number of hypotheses tested. If R is simply a vector of zeros with a one in the k -th element, for

²For $\mathbf{X} \sim N(\mathbf{0}, \mathbf{I}_n)$, $\mathbf{X}'\mathbf{A}\mathbf{X}$ is χ^2 with degrees of freedom equal to the trace of A when A is idempotent and symmetric. Independence happens because $\hat{\beta}$ and $\hat{\mathbf{e}}$ are independent.

³Since $\hat{\mathbf{e}}'\hat{\mathbf{e}}/\sigma^2$ is $\chi^2(N - K)$, it has mean $N - K$ and variance $2(N - K)$. Dividing by $N - K$ produces a random variable with mean 1 and variance converging to zero as $N \rightarrow \infty$, implying $t_k \rightarrow z_k$.

example, we are back to the case of testing a hypothesis about a single element of β . In general, however, R can encode a variety of linearly independent hypotheses about elements of β .

Under the assumption of the normal regression model, the **F-stat** or **F-ratio** is a useful way to test the restrictions. This test statistic is given by:

$$F = \frac{(R\hat{\beta} - r)'[R(\mathbf{X}'\mathbf{X})^{-1}R']^{-1}(R\hat{\beta} - r)/m}{\hat{\sigma}^2} \\ = (R\hat{\beta} - r)'[\widehat{RVar}(\hat{\beta}|\mathbf{X})R']^{-1}(R\hat{\beta} - r)/m$$

This test stat is distributed $F(m, N - K)$. To understand why, recall that if $Y = (X_1/k_1)/(X_2/k_2)$ where X_1 and X_2 are χ^2 with degrees of freedom k_1 and k_2 , then Y is distributed $F(k_1, k_2)$. Under the null, $R\hat{\beta} - r$ is normally distributed with variance $\sigma^2 R(\mathbf{X}'\mathbf{X})^{-1}R'$ (because normals are closed under linear transformations). When R has full row rank (so no redundant hypotheses), this matrix will be invertible. Multiplying the top and bottom of F by $1/\sigma^2$ gives:

$$F = \frac{(R\hat{\beta} - r)'[\sigma^2 R(\mathbf{X}'\mathbf{X})^{-1}R']^{-1}(R\hat{\beta} - r)/m}{\hat{\sigma}^2/\sigma^2}$$

which as promised is just the ratio of two independent χ^2 s (independence will be shown in your homework). F-stats can be evaluated against the critical values from the right tail of the $F(k_1, k_2)$ distribution. Bigger test stats imply more rejection. Critical values are decreasing in m , and for large N the 5% critical values is below 2 when $m \geq 7$.

Notice that when R involves only a single restriction on a single coefficient β_k , $F = t_k^2$. In this sense the F-test nests t-tests. Notice also that as N becomes very large, the $F(m, N - k)$ will converge to a $\chi^2(m)$ distribution, consistent with convergence of t-stats to z-scores.

Regression inference asymptotically

This is all beautiful, but only worked under the very strong assumption of normality. Ideally, we'd like to rely on a weaker assumptions to conduct inference that remains informative. Thankfully, we can apply central limit theorem results to justify normality as an *approximation* when the data are sufficiently large. To see why this is justified, recall that the $\sqrt{N}$ -rescaled and properly centered OLS estimator can be written:

$$\sqrt{N}(\hat{\beta} - \beta) = \sqrt{N}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{U} = \left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} \sqrt{N}\frac{\mathbf{X}'\mathbf{U}}{N}$$

We have already discussed how $(\mathbf{X}'\mathbf{X}/N)^{-1}$ converges to $E[XX']^{-1}$ by the law of large numbers. The second term is simply:

$$\sqrt{N}\frac{\mathbf{X}'\mathbf{U}}{N} = \sqrt{N}\left(\frac{1}{N}\sum_{i=1}^N X_i U_i\right)$$

which by the standard central limit theorem converges to a $N(0, Var(X_i U_i))$ distribution whenever that variance is finite. Putting the two results together implies that:

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, E[XX']^{-1}Var(XU)E[XX']^{-1})$$

Thus, regardless of the underlying distributions of Y and X , under iid sampling $\hat{\beta}$ has a limiting normal distribution provided variances are finite and $E[XX']$ is invertible. When N is sufficiently large, we may be comfortable assuming this approximation is “good enough” that we can treat $\hat{\beta}$ as *actually* normally distributed and conduct testing / inference as we did above.

To make this useful, however, we need an estimator of $\Sigma = E[XX']^{-1}Var(XU)E[XX']^{-1} = E[XX']^{-1}E[XX'U^2]E[XX']^{-1}$. As usual, the natural choice is the simple “plug-in” estimator:

$$\hat{\Sigma} = \left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1} \left(\frac{\mathbf{X}'\mathbf{D}\mathbf{X}}{N} \right) \left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1}$$

where $\mathbf{D}$ is an N -by- N diagonal matrix with $\hat{U}_i^2$ on the i th diagonal (make sure you understand the matrix math for the “meat” of this sandwich formula). This estimator is known as the **Eicker-[Huber-White](#)** (EHW) or **robust** variance estimator, since it allows for arbitrary **heteroskedasticity**. That is, we haven’t imposed that $E[U^2|X] = \sigma^2$. If we did impose homoskedasticity, the variance formula would simplify significantly, since:

$$E[XX']^{-1}E[XX'U^2]E[XX']^{-1} = E[XX']^{-1}E[XX'E[U^2|X]]E[XX']^{-1} = \sigma^2 E[XX']^{-1}$$

You’ll notice that in the homoskedastic case, the variances is identical to the average variance in the normal model studied earlier. It has the natural estimator $(\mathbf{X}'\mathbf{X})^{-1}\hat{\mathbf{U}}'\hat{\mathbf{U}}\frac{N}{N-k}$. One can show that the EHW estimator is consistent regardless of whether Y is hetero or homoskedastic, while the homoskedastic estimator is consistent under homoskedasticity.

Keep in mind that in both the homoskedastic and heteroskedastic case, we are estimating the *asymptotic* variance of $\hat{\beta}$. That is, the variance of $\sqrt{N}\hat{\beta}$. To get the sampling variance of $\hat{\beta}$, we need to divide by $\sqrt{N}$. By the usual properties of variances, this requires multiplying by the asymptotic variance by $1/N$, so that the EHW robust estimator for the *sampling variance* is simply:

$$\hat{V}_{\hat{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{D}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$$

Excellent! We have invoked the CLT to justify a normal approximation and constructed an estimator for the sampling variance of $\hat{\beta}$. Even though this approximation only becomes good when N is very large, it is still customary to use the same finite sample terminology for testing and inference as in the normal regression model studied above. That is, researchers typically report what they call a t-stat instead of a z-score, even though normality only kicks in at large N , after which t-stats effectively become z-scores (they are close after $N = 100$ or so). Technically, the asymptotic approximation only guarantees that the t-stat is asymptotically standard normal and nothing about the finite-sample distribution, so some people prefer to use the $N(0, 1)$ distribution to compute critical values rather than the $t(N - K)$ distribution. As noted earlier, using the t distribution as the reference distribution instead is going to be more conservative.

Along the way, we also relaxed the homoskedasticity assumption. These days, we almost never use the homoskedastic estimator for sampling variances because we rarely have good reasons to believe we know something about the conditional variance of Y . The robust estimator also nests the homoskedastic case, so it is consistent either way. Assuming homoskedasticity can lead to misleading inference when the assumption is violated. In the spirit of being statistically agnostic,

it is therefore common to use “robust” standard errors whenever possible. In Stata, this is easy to accomplish using the `, robust` option after `reg`. In Python’s statsmodels package, you can ask for the EHW variance formula by passing an argument to `fit`, e.g., `model_ols.fit(cov_type=“HCO”)`.

Allowing for heteroskedasticity does not change how one would compute a t-stat aside from the standard error used in the denominator, but it does affect testing multiple linear hypotheses. In the normal model, we relied on homoskedasticity to derive the sampling distribution of the F-statistic, since we needed to multiply and divided by the common σ^2 . Under heteroskedasticity, there is no single σ^2 to divide by, so we instead rely more heavily on the asymptotic approximations and construct a [Wald statistic](#):

$$W = (R\hat{\beta} - r)'[R\widehat{Var}(\hat{\beta})R']^{-1}(R\hat{\beta} - r)$$

Under the null that $R\hat{\beta} = r$, W will converge to a $\chi^2(m)$ distribution.⁴ We can use the critical values of this distribution to define a testing procedure. As with F-stats, large Walds imply rejection.

Notice that in one sense the Wald is m times the limiting form of the F stat when written as $(R\hat{\beta} - r)'[R\widehat{Var}(\hat{\beta})R']^{-1}(R\hat{\beta} - r)/m$. You should have expected this, since as discussed in our probability review $k_1 F_{k_1, \infty}$ is a $\chi^2(k_1)$ distribution.

Despite the fact that the F test is inappropriate under heteroskedasticity, some popular software packages like Stata will still report F tests for linear hypotheses after using `, robust`. Under the hood, Stata is simply reporting W/m and using the $F(m, N - k)$ distribution to compute critical values. I find it more conceptually coherent when invoking asymptotic approximations and a heteroskedastic model to simply use a Wald test, although some would argue using the F as the reference distribution is a prudent finite sample adjustment. It matters a lot which one you use, you probably don’t have enough data to justify your asymptotic approximations anyways. Later we will discuss the use of “robust” F statistics as a diagnostic for a separate issue: weak instruments.

Walds can also be used to test certain [non-linear hypotheses](#) using the delta method. Suppose we wish to know whether:

$$g(\beta) = 0$$

where a is a differentiable vector-valued function. This function could return, for example, the product of two elements of β or any other non-linear combination of elements of β . Since $\sqrt{N}\hat{\beta} \xrightarrow{d} N(\beta, \Sigma)$, delta method logic tells us that $\sqrt{N}g(\hat{\beta}) \xrightarrow{d} N(g(\beta), G(\beta)\Sigma G(\beta)')$ where G is the Jacobian. We can thus construct a Wald test of $g(\beta) = 0$ using:

$$W = g(\hat{\beta})'[G(\hat{\beta})\widehat{Var}(\hat{\beta})G(\hat{\beta})']^{-1}g(\hat{\beta})$$

which will be asymptotically $\chi^2(m)$, where m is the dimension of $g(\beta)$. For this test to be well defined we require that $G(\hat{\beta})$ has full row rank, so that that no restriction are redundant. You

⁴Note that $Var(\hat{\beta})^{-1/2}(\hat{\beta} - \beta)$ is asymptotically $N(0, \mathbf{I}_k)$, since for a normal random variable X with covariance matrix Σ , $Var(\Sigma^{-1/2}X) = \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = \mathbf{I}$ (recall that $\Sigma^{-1/2}\Sigma\Sigma^{-1/2}(\Sigma^{-1}\Sigma) = \Sigma^{-1/2}\Sigma^{-1/2}\Sigma = \mathbf{I}$). Because $R\hat{\beta}$ has variance $RVar(\hat{\beta})R'$, $(RVar(\hat{\beta})R')^{-1/2}R\hat{\beta}$ is also $N(0, \mathbf{I}_m)$. Quadratic forms of m independent normals are $\chi^2(m)$.

can also see that this non-linear case nests the linear case we discussed above if we simply take $g(\beta) = R\beta - r$.

Another issue that arises when allowing for heteroskedasticity is the potential bias of our estimator for the asymptotic variance for $\hat{\beta}$. Bias could be a big deal if it leads us to systematically under or over reject null hypothesis. It is easy to see that the homoskedastic estimator $(\mathbf{X}'\mathbf{X})^{-1}\hat{\mathbf{U}}'\hat{\mathbf{U}}\frac{N}{N-k}$ is conditionally unbiased (and therefore unconditionally unbiased as well), since as we showed above $\hat{\mathbf{U}}'\hat{\mathbf{U}}/(N-k)$ is unbiased for σ^2 .

It is more difficult to show when the EHW variance estimator $\hat{V}_{\hat{\beta}}$ is unbiased under heteroskedasticity. The EHW estimator can be written as:

$$\hat{V}_{\hat{\beta}} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i X_i' \hat{U}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

As we discussed above, $\hat{U}_i^2$ is in general *too small* due to how greedy OLS is about minimizing squared errors in sample. The homoskedastic estimator takes care of this by adjusting by $N-K$, so it is common to do the same for $\hat{V}_{\hat{\beta}}$ by defining:

$$\hat{V}_{\hat{\beta}}^r = \frac{N}{N-K} (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i X_i' \hat{U}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1} = \frac{N}{N-K} \hat{V}_{\hat{\beta}}$$

Stata will do this by default when you use `, robust`. This approach is a little ad-hoc: there is no general theory as to why it should work well. It is not equivalent to the homoskedastic estimator under homoskedasticity except in special cases.

An alternative is try to construct estimators $\hat{U}_i^2$ that don't suffer from the over-fitting issue. One way to do that is to compute $\hat{U}_i$ using an estimator of β that excludes i , i.e., a leave-one-out estimator motivated by ideas from cross-validation:

$$\begin{aligned} \tilde{U}_i &= Y_i - X_i' \hat{\beta}_{-i} \\ \hat{\beta}_{-i} &= \left(\sum_{j \neq i} X_j X_j' \right)^{-1} \left(\sum_{j \neq i} X_j Y_j \right) \end{aligned}$$

This looks like a pain because you need to compute a different $\hat{\beta}$ for every observation i , but it turns out that a more convenient expression for $\tilde{U}_i$ can be computed using observation i 's [leverage value](#), defined as $h_{ii} = X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i$. One can show that leverage values fall between 0 and 1 and that h_{ii} is the i th diagonal element of $\mathbf{P}_{\mathbf{X}}$, which implies that $\sum_{i=1}^N h_{ii} = \text{trace}(\mathbf{P}_{\mathbf{X}}) = K$. Leverage values summarize how much influence observation i has over $\hat{\beta}$ and hence $X_i'\hat{\beta}$. The intuition is that because $\hat{\mathbf{Y}} = \mathbf{P}_{\mathbf{X}}\mathbf{Y}$, we can write $\hat{Y}_i$ as the i th row of $\mathbf{P}_{\mathbf{X}}$ times $\mathbf{Y}$. Observation i 's contribution is the i th element of that row, so the i th diagonal of $\mathbf{P}_{\mathbf{X}}$. Observations with high h_{ii} are said to have high leverage.

It turns out that (I leave the proof as a somewhat tedious [exercise](#)):

$$\hat{\beta}_{-i} = \hat{\beta} - (1 - h_{ii})^{-1}(\mathbf{X}'\mathbf{X})^{-1}X_i\hat{U}_i$$

Substituting for $\hat{\beta}_{-i}$ in the above, this implies that what are known as [prediction errors](#) or [out-of-sample residuals](#) can be computed as:

$$\begin{aligned}\tilde{U}_i &= Y_i - X_i' \hat{\beta} + X_i'(1 - h_{ii})^{-1}(\mathbf{X}'\mathbf{X})^{-1}X_i\hat{U}_i \\ &= \hat{U}_i + (1 - h_{ii})^{-1}h_{ii}\hat{U}_i = (1 - h_{ii})^{-1}\hat{U}_i\end{aligned}$$

An alternative variance estimator using $\tilde{U}$ would be given by:

$$\begin{aligned}\hat{V}_{\hat{\beta}}^{HC3} &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i X_i' \tilde{U}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i X_i' (1 - h_{ii})^{-2} \hat{U}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

I call this *HC3* because it is what Stata will give you if you ask for , *vce(hc3)*.

We showed earlier that $E[\hat{U}'\hat{U}|\mathbf{X}] = E[\mathbf{U}'\mathbf{M}_\mathbf{X}\mathbf{U}|\mathbf{X}]$, which simplifies to $\sigma^2 \text{trace}(\mathbf{M}_\mathbf{X})$ under conditional homoskedasticity. This implies that conditional variance of the i th residual is given by $\text{Var}(\hat{U}_i|\mathbf{X}) = (1 - h_{ii})\sigma^2$, again under conditional homoskedasticity (replace $\hat{U}$ with the zeros except in the i th element). Since h_{ii} depends on i , these residuals are conditionally heteroskedastic despite the fact that $\text{Var}(U_i|\mathbf{X}) = \sigma^2$. A third and final approach to consistent variance estimation is to use [standardized residuals](#) that undo this heteroskedasticity using:

$$\hat{V}_{\hat{\beta}}^{HC2} = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^N X_i X_i' (1 - h_{ii})^{-1} \hat{U}_i^2 \right) (\mathbf{X}'\mathbf{X})^{-1}$$

which is what Stata gives you when using , *vce(hc2)*.

Because $(1 - h_{ii})^{-2} > (1 - h_{ii})^{-1} > 1$, $\hat{V}_{\hat{\beta}}^{HC3} > \hat{V}_{\hat{\beta}}^{HC2} > \hat{V}_{\hat{\beta}}$ in a matrix sense, so using HC3 is going to lead to more conservative inference than HC2, and using HC2 is going to lead to more conservative inference than the unadjusted EHW variance estimator. In a world where the data really is homoskedastic, the homoskedastic estimator will of course be unbiased. Because $E[\hat{U}_i^2|\mathbf{X}] = (1 - h_{ii})\sigma^2$ in a homoskedastic world, however, HC2 will be unbiased as well, while HC3 will generally be too large. There is a long debate about what variance estimator to use when we are in a heteroskedastic setting (see e.g. Davidson and MacKinnon, 1993). My advice is to use at the very least $\hat{V}_{\hat{\beta}}^r$, HC2, or HC3, with the latter two likely preferred given their stronger theoretical grounding. Some simulation studies suggest HC3 is the best choice.

There a broad range of other settings where *none* of these variance estimators will be appropriate because we have broken parts of the iid sampling assumption. We will leave that interesting but thorny issue alone for now and return to it later in the course.

As a final point, its worth coming back to our discussion of FWL. For the linear regression $Y = X'\beta + W'\gamma + \hat{U}$, are the standard errors for $\hat{\beta}$ one gets from regressing $\tilde{Y}$ on $\tilde{X}$ generally the same as what one would get from a regression of Y on both X and W ? A standard variance estimator for the residualized regression would be:

$$\hat{V}_{\hat{\beta}} = (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}(\tilde{\mathbf{X}}'\mathbf{D}\tilde{\mathbf{X}})(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}$$

with $\mathbf{D} = \text{diag}(\hat{U}_i^2)$ as before.

Is this estimator identical to the submatrix of the full variance estimator corresponding to β ? The answer is yes, and exactly so, when we are considering $\hat{V}_{\hat{\beta}}$, although it takes some annoying algebra to show it. $\hat{V}_{\hat{\beta}}^r$ is also equivalent after accounting for degrees of freedom due of the regressors that were residualized off. In general, however HC2 and HC3 will not be equivalent (see Ding 2021), so if you're a stickler for particular variance estimators working with the residualized regressors and outcomes may not be sufficient.

Optimality

We have just discussed how to quantify uncertainty about β based on the standard OLS estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$. A natural question is whether this estimator is the “best” estimator of β . Is there another choice that would have smaller asymptotic variance? If so, we might prefer it, since for any given N our standard errors are likely to be smaller.

One of the most celebrated results in classical regression theory, the [Gauss-Markov theorem](#), gives precise conditions under which OLS is the best choice. It says that if the conditional expectation of Y is linear in X , so that $E[U|X] = 0$, and U is homoskedastic, so that $E[U^2|X] = \sigma^2$, then OLS is the minimum variance unbiased linear estimator. Linear here means that the estimator can be written as $\tilde{\beta} = \mathbf{A}'\mathbf{Y}$, with $\mathbf{A}$ determined by $\mathbf{X}$ (i.e., $\mathbf{A}(\mathbf{X})$).

To see why, observe that if $\tilde{\beta} = \mathbf{A}'\mathbf{Y}$ is unbiased, then:

$$E[\tilde{\beta}|\mathbf{X}] = \mathbf{A}'E[\mathbf{Y}|\mathbf{X}] = \mathbf{A}'\mathbf{X}\beta$$

This implies unbiasedness requires $\mathbf{A}'\mathbf{X} = \mathbf{I}_k$, the identity matrix. Homoskedasticity implies that:

$$\begin{aligned} \text{Var}(\mathbf{A}'\mathbf{Y}|\mathbf{X}) &= \text{Var}(\mathbf{A}'(\mathbf{X}\beta + \mathbf{U})|\mathbf{X}) = \text{Var}(\mathbf{A}'\mathbf{U}|\mathbf{X}) \\ &= E[\mathbf{A}'\mathbf{D}\mathbf{A}|\mathbf{X}] = \mathbf{A}'\mathbf{A}\sigma^2 \end{aligned}$$

where $\mathbf{D} = E[\mathbf{U}\mathbf{U}'|\mathbf{X}]$. We know that the OLS choice of $\mathbf{A}$ is $\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$, with $\mathbf{A}'\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}$. Suppose that we used some other $\mathbf{A}$ satisfying $\mathbf{A}'\mathbf{X} = \mathbf{I}_k$. Then:

$$\begin{aligned} \mathbf{A}'\mathbf{A} - (\mathbf{X}'\mathbf{X})^{-1} &= (\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})'(\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) - (\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})'(\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) \\ &\quad + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) + (\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1})'(\mathbf{A} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) = \mathbf{C}'\mathbf{C} \end{aligned}$$

where the last line follows because $\mathbf{X}'\mathbf{A} = \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{I}_k$. The matrix $\mathbf{C}'\mathbf{C}$ is positive definite, meaning $\mathbf{c}'\mathbf{C}'\mathbf{C}\mathbf{c} > 0$ for $\mathbf{c} \neq 0$.⁵ Thus any choice for $\mathbf{A}$ besides $\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ is going to lead to a larger variance, because the difference between $\text{Var}(\mathbf{A}'\mathbf{Y})$ and $\text{Var}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y})$ will be positive definite. This applies to any element of β or linear combinations thereof.

⁵ $\mathbf{A} = \mathbf{G}'\mathbf{G}$ is positive semi-definite and positive definite when $\mathbf{G}$ has full rank. Recall that $\mathbf{c}\mathbf{A}\mathbf{c} = \mathbf{c}\mathbf{G}'\mathbf{G}\mathbf{c}$ is simply the inner product of the vector $\mathbf{a} = \mathbf{G}\mathbf{c}$.

Note that this result does not preclude some other *biased* estimator achieving lower variance. This is in fact trivial, since we can consider $\hat{\beta} = 0$ which is likely to be very biased but has zero variance. Likewise, Gauss-Markov does not imply there is no alternative *non-linear* estimator that outperforms OLS.

The proof of the Gauss-Markov theorem also provides some hints about how to think about efficiency in cases where the model is *not* homoskedastic. Consider the [generalized least squares](#) estimator given by:

$$\hat{\beta}_{GLS} = (\mathbf{X}'\mathbf{D}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}^{-1}\mathbf{Y}$$

The conditional variance of this estimator is:

$$\begin{aligned} \text{Var}(\hat{\beta}_{GLS}|\mathbf{X}) &= (\mathbf{X}'\mathbf{D}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}^{-1}\text{Var}(\mathbf{Y}|\mathbf{X})\mathbf{D}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{D}^{-1}\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{D}^{-1}\mathbf{X})^{-1}\mathbf{I}_k \end{aligned}$$

Notice that of course this simplifies back to the homoskedastic variance formula when $\mathbf{D} = \sigma^2\mathbf{I}_k$. This is the best unbiased linear estimator when we assume that Y is linear in X but do not impose homoskedasticity.

To see what is going on here, define $\mathbf{C}$ such that $\mathbf{C}'\mathbf{C} = \mathbf{D}^{-1}$. Then we can write $\hat{\beta}_{GLS}$ as

$$\begin{aligned} \hat{\beta}_{GLS} &= (\mathbf{X}'\mathbf{C}'\mathbf{C}\mathbf{X})^{-1}\mathbf{X}'\mathbf{C}'\mathbf{C}\mathbf{Y} \\ &= (\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{Y}} \end{aligned}$$

where $\tilde{\mathbf{X}} = \mathbf{C}\mathbf{X}$ and $\tilde{\mathbf{Y}} = \mathbf{C}\mathbf{Y}$. So this is just regular OLS with the transformed regressors $\tilde{\mathbf{X}}$ and transformed outcome $\tilde{\mathbf{Y}}$. The residuals will be given by $\tilde{\mathbf{U}} = \mathbf{C}\mathbf{U}$ and have variance:

$$E[\tilde{\mathbf{U}}\tilde{\mathbf{U}}'|\mathbf{X}] = \mathbf{I}_k$$

In other words, the transformed regression is homoskedastic, so regular Gauss-Markov will apply.

How should you interpret these transformed regressors? Under independent sampling, $\mathbf{D}$ is diagonal with typical element given by $\sigma_i^2 = E[U_i^2]$. The matrix $\mathbf{C}$ will also be diagonal, with typical element $1/\sigma_i$. This means we can re-write the estimator in summation form as:

$$\begin{aligned} \hat{\beta}_{GLS} &= \left(\sum_{i=1}^N \frac{X_i X_i'}{\sigma_i^2} \right)^{-1} \left(\sum_{i=1}^N \frac{X_i Y_i}{\sigma_i^2} \right) \\ &= \left(\sum_{i=1}^N w_i X_i X_i' \right)^{-1} \left(\sum_{i=1}^N w_i X_i Y_i \right) \\ w_i &= \frac{\sigma_i^{-2}}{\sum_{j=1}^N \sigma_j^{-2}} \end{aligned}$$

GLS is thus just a type of [weighted regression](#), where the weights sum to one and are proportional the inverse conditional variance of Y_i . Observations with higher conditional variance get less weight because we expect more noise in Y_i , and thus more uncertainty about β .

Of course $\hat{\beta}_{GLS}$ is an [infeasible](#) estimator because the conditional variances σ_i^2 are usually unknown (there are some special cases where this might not be true). [Feasible generalized least squares](#) replaces $\mathbf{D}$ with a consistent estimator and then forges ahead. Doing this requires having a model for $E[U_i^2|X_i]$, which is sometimes assumed be linear in a subset of the covariates Z_i so that $E[U_i^2|X_i] = Z_i'\gamma$. If γ is unknown, it can be consistently estimated using a regression of $\hat{U}_i^2$ on Z_i and we can compute the plug-in version of FGLS using $\hat{\gamma}$. It turns out that if the assumption on the form of conditional variance is correct, FGLS is asymptotically equivalent to GLS (same asymptotic variance). Finite sample performance is another story.

As you can imagine, FGLS is not used too much in practice because we are typically unwilling to take a stand on the form of conditional heteroskedasticity. Nevertheless, related ideas come up frequently, particularly when dealing with panel data or data that captures group averages. If X_i and Y_i are averages from groups of size n_i , then we naturally expect less variance for observations where n_i is large. Weighting regressions by group size helps account for this in the hope that the resulting estimator is more efficient.